

# Understanding Autoencoders: How Neural Networks Learn to Reconstruct Images

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**GitHub Link: <https://github.com/Lakshman7257/MLNN-Assingment>**

## 1. Introduction

### 1.1 Reconstruction as opposed to classification

Numerous neural network tutorials commence with classification. In categorization, an image is presented to a model, which then predicts a label such as shirt, shoe, bag, or outfit. An autoencoder operates differently. It does not acquire the ability to forecast a class label. Rather, it acquires the ability to reconstitute its own input.

This lesson demonstrates the process of inputting a Fashion-MNIST image into the model, compressing it into a more compact internal representation, and subsequently reconstructing it as an output image. The input image and the target image are identical. The autoencoder is beneficial for comprehending picture compression, reconstruction, and feature extraction.

The primary inquiry in this tutorial is:

**What information can a neural network retain when an image is processed through a narrow bottleneck?**

This inquiry is significant due to the bottleneck's inability to accurately store every pixel. The model must ascertain which visual information is most essential for image reconstruction.

## 2. Architecture of Autoencoders

### 2.1 Encoder

The encoder constitutes the initial component of the autoencoder. It acquires the image and compresses it into a reduced set of values. Each Fashion-MNIST image in this lesson measures  $28 \times 28$  pixels. Upon flattening, a single image yields 784 pixel values. The encoder compresses this 784-dimensional input into a more compact representation.

The compression process is the fundamental distinction between merely duplicating an image and acquiring a valuable representation. The encoder must convert the input into a more concise format that retains sufficient information for reconstruction.

### 2.2 Constraint

The constraint lies in the compressed central layer of the autoencoder. It is diminutive compared to the original input, hence establishing an information constraint. A beneficial approach to comprehend it is to envision it as a slender bridge. The original image comprises 784 pixel values, however the bottleneck may have just 8, 32, or 64 values. As not all details can traverse this confined space, the model must retain the most pertinent structure.

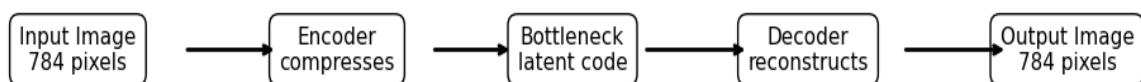
This constitutes the principal concept of the tutorial. A narrower bottleneck typically results in enhanced compression, albeit at the potential cost of greater detail loss. A greater bottleneck can retain more information, although the enhancement may diminish beyond a certain threshold.

## 2.3 Decoder

The decoder constitutes the second component of the autoencoder. It acquires the bottleneck representation and strives to reconstruct the original image. The output matches the input size, indicating that the decoder generates 784 reconstructed pixel values. If the reconstruction resembles the original image, the model has acquired a valuable compressed representation.

Figure 2: Fundamental autoencoder architecture illustrating the compression of an image by the encoder, depicted in the bottleneck, and its reconstruction by the decoder.

Figure 2: Basic Autoencoder Structure



A flow diagram illustrating an input image traversing through an encoder, bottleneck, and decoder to generate a reconstructed output image.

## 3. Mathematical Concept

### 3.1 Encoding and Decoding

The encoder may be depicted as:

$$z = f_{\theta}(x)$$

$$\hat{x} = g_{\phi}(z)$$

Here,  $g_{\phi}$ . The decoder is represented, and  $\hat{x}$  denotes the reconstructed picture.

### 3.2 Reconstruction Loss

The model acquires knowledge by minimizing the disparity between the original image and the reconstructed image. This tutorial employs mean squared error.

$$L(x, \hat{x}) = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

## 4. Dataset Preparation

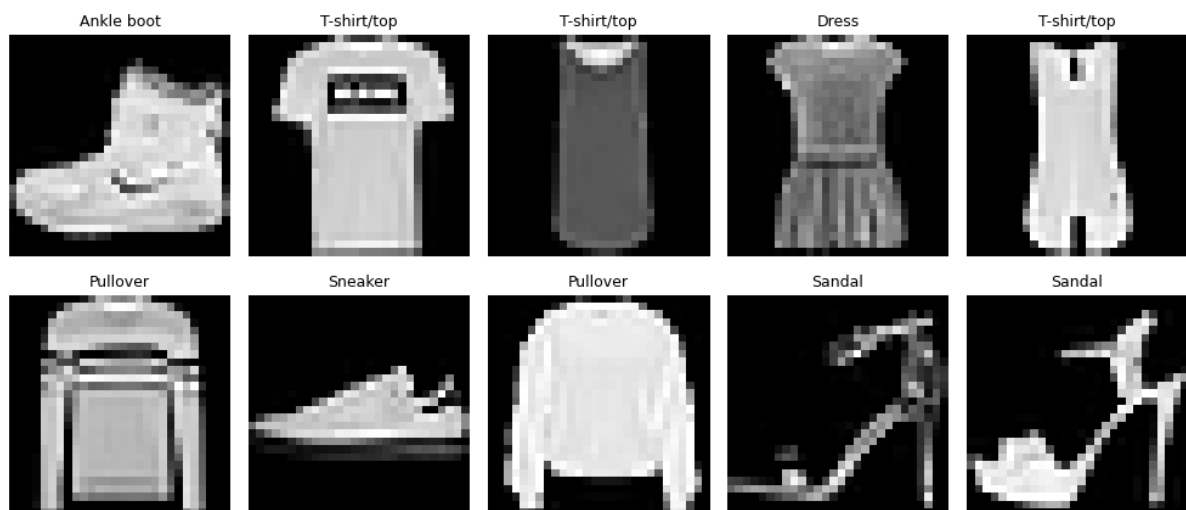
### 4.1 Fashion-MNIST

Fashion-MNIST comprises grayscale photos of apparel, including shirts, shoes, purses, gowns, and sandals. The class labels are accessible, yet they are not utilized for training the autoencoder. The model acquires knowledge directly from the photos.

The pixel values are normalized from 0–255 to 0–1. Each  $28 \times 28$  image is subsequently flattened into 784 values for input into a dense autoencoder.

Figure 1: Illustrative Fashion-MNIST photos utilized in the autoencoder tutorial.

Figure 1: Example Fashion-MNIST Images



Alt text: A grid of grayscale Fashion-MNIST apparel photos, with ankle boots, blouses, dresses, pullovers, sneakers, and sandals.

# Constructing the Autoencoder

## Components of the Model

The autoencoder implemented in the notebook is deliberately simplistic to ensure the primary notion is evident. The input layer accepts a flattened Fashion-MNIST image with 784 pixel values. The encoder compresses this input via a dense layer comprising 128 units. The bottleneck subsequently condenses the representation into 32 values. Subsequently, the decoder amplifies the bottleneck representation via an additional dense layer, culminating in the final output layer that generates 784 reconstructed pixel values.

| Component        | Role                                                |
|------------------|-----------------------------------------------------|
| Input layer      | Receives the flattened 784-pixel image.             |
| Encoder layer    | Compresses the image into a smaller representation. |
| Bottleneck layer | Stores the compact latent code.                     |
| Decoder layer    | Expands the latent code for reconstruction.         |
| Output layer     | Produces the reconstructed 784-pixel image.         |

This framework is advantageous for instruction as each component has a distinct function. The encoder compresses data, the bottleneck constrains information, and the decoder reconstructs it.

## Objectives of the Model Training

The primary distinction is that the autoencoder undergoes training in a manner distinct from that of a classifier. In a classifier, the input constitutes an image, while the target represents a class label. In this autoencoder, the input and output are identical images. The model is presented with an image and tasked with reconstructing it as accurately as possible.

This indicates that the training pair is:

$$x \rightarrow x$$

# Analyzing the Reconstruction Loss

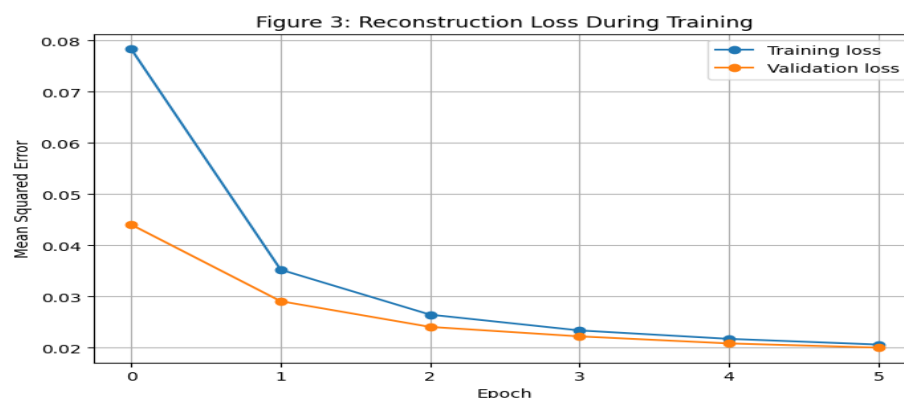
## Interpretation of the Loss Curve

Upon training the autoencoder, the loss curve illustrates the variation of reconstruction error over time. The training loss in the notebook output diminished

from approximately 0.0784 in the first epoch to roughly 0.0206 by the sixth epoch. The validation loss diminished from approximately 0.0441 to roughly 0.0200.

This pattern indicates that the model enhanced its reconstructions throughout the training process. The initial epoch has significant inaccuracy as the model has not yet acquired the ability to effectively compress and reconstruct the images. During the training process, the weights of the encoder and decoder are adjusted, resulting in reconstructed images that increasingly resemble the original images.

Figure 3: Reconstruction loss for training and validation throughout six epochs.



Alt text: A line chart depicting the decline of training and validation mean squared error throughout the autoencoder training process.

### Interpretation of the Curve

The primary focus for interpreting Figure 3 is not solely the precise loss value. The configuration of the curve is significant. A declining curve indicates that the model is acquiring knowledge. The validation loss diminishes and stays around the training loss, indicating that the model is not only memorizing the training images. It can even rebuild previously unseen validation images with considerable accuracy.

In this tutorial, the loss curve serves as a metric for evaluating learning progress. Prior to examining the reconstructed images, it is confirmed that the model has diminished the disparity between the original and rebuilt pixels.

### A Comparison of Original and Reconstructed Images

#### Criteria for Evaluation

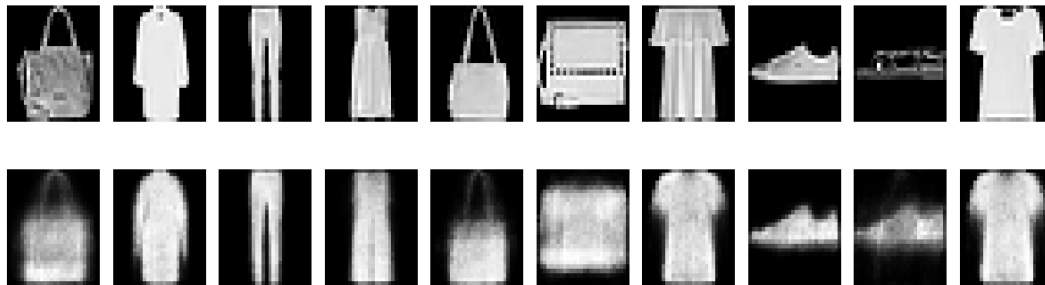
The subsequent step is to examine the photographs directly. Figure 4 juxtaposes original Fashion-MNIST photos with their respective reconstructed outputs from the autoencoder. This is the most comprehensible method to grasp the concept of rebuilding.

Contrast the upper row with the lower row. The rebuilt images maintain the overall form of the objects. Handbags retain their characteristic appearance, footwear

remains distinguishable, and the contours of garments are still apparent. Nonetheless, the reconstructions have a softer quality and diminished detail compared to the originals. Refined edges, textures, and distinct borders are somewhat obscured.

Figure 4: Original Fashion-MNIST images juxtaposed with their reconstructed outputs from the autoencoder.

Figure 4: Original Images and Autoencoder Reconstructions



alt text: An picture grid consisting of two rows: the first row displays original apparel photographs, while the second row presents their recreated counterparts.

## 7.2 The Imperfections of Reconstructions

The reconstructions are not precise replicas due to the necessity of the image traversing the bottleneck. The original image comprises 784 pixel values, although the primary model condenses it to merely 32 values prior to reconstruction. This constrained space cannot retain every minor nuance. The model learns to retain the most essential visual structure required to reconstruct a recognizable version of the image.

The primary lesson is that an autoencoder does not merely replicate a picture. It acquires a compressed representation that retains sufficient information for reconstruction.

## Bottleneck Experiment

### What is the rationale for comparing bottleneck sizes?

The bottleneck is the crucial component of this tutorial as it regulates the volume of information the autoencoder transmits from the encoder to the decoder. The notebook trains three autoencoders with bottleneck sizes of 8, 32, and 64 to enhance visibility. Each model is presented with identical Fashion-MNIST photos and endeavors to recreate them. The sole significant alteration is the dimension of the latent code.

An 8-value bottleneck necessitates significant compression. The model should retain only the most general visual framework, resulting in the potential removal of minor

details. A bottleneck of 32 values provides the model with more capacity to maintain shape and contour. A bottleneck of 64 values provides more capacity, although the enhancement may diminish as a significant portion of the essential structure may already be represented at 32 values.

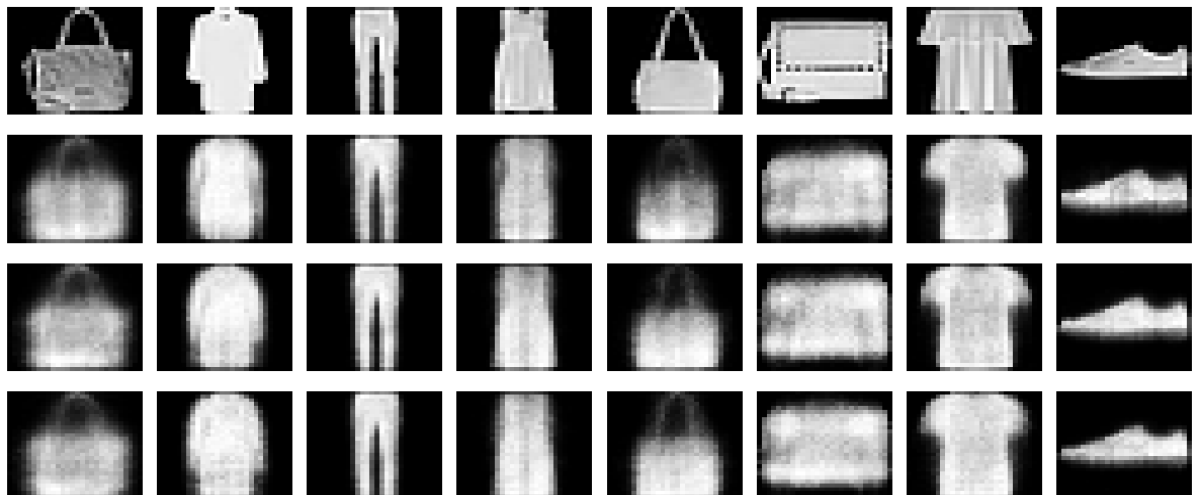
### Analyzing the visual comparison

Figure 5 juxtaposes the original images with reconstructions derived from the three bottleneck dimensions. The row with a latent size of 8 should be prioritized for inspection. These reconstructions typically maintain the overarching item category, although they exhibit a softer and less detailed appearance. This indicates that the model has grasped the primary structure but is unable to preserve all pixel-level details due to the limited representation.

The latent 32 row typically yields more distinct reconstructions. Apparel silhouettes and footwear contours become increasingly identifiable. The latent 64 row may exhibit marginally more sharpness, however it is not significantly distinct from latent 32. Enhancing the bottleneck size can augment reconstruction; however, the enhancement is not invariably proportional.

Figure 5: Impact of bottleneck dimensions on reconstruction quality. Smaller bottlenecks compress more intensely and result in more loss of visual detail.

Figure 5: Effect of Bottleneck Size on Reconstruction



A grid juxtaposing genuine Fashion-MNIST photos with reconstructions from autoencoders utilizing bottleneck sizes of 8, 32, and 64.

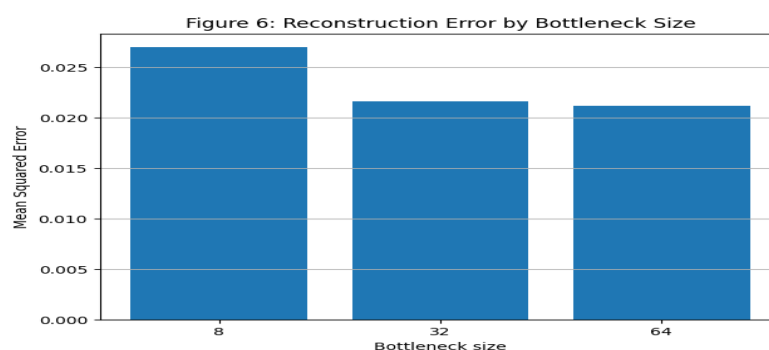
### Reconstruction Error

The reconstruction mistake facilitates visual comparison. The model with a bottleneck size of 8 had the largest mean squared error, whereas bottleneck values of 32 and 64 resulted in reduced errors.

| Bottleneck size | Mean squared error |
|-----------------|--------------------|
| 8               | 0.02694            |
| 32              | 0.02161            |
| 64              | 0.02116            |

The most significant enhancement transpires between 8 and 32. Augmenting from 32 to 64 yields merely a little further reduction. This illustrates the trade-off between compression and overall reconstruction quality.

Figure 6: Average squared reconstruction error for bottleneck sizes 8, 32, and 64.



A bar chart illustrating reconstruction error over three bottleneck sizes, indicating the greatest error for a bottleneck size of 8.

## Practical Applications and Constraints

Autoencoders are advantageous for dimensionality reduction, feature extraction, image compression, noise reduction, and anomaly detection. A model designed to recreate standard images may exhibit increased reconstruction error for atypical images, hence facilitating anomaly detection. Nonetheless, reconstruction does not invariably equate to genuine comprehension. An excessively big bottleneck may enable the model to replicate inputs with undue ease, whereas an excessively small bottleneck may eliminate crucial information. Dense autoencoders also overlook a significant portion of the spatial structure that convolutional autoencoders can leverage.

## Conclusion

An autoencoder acquires knowledge by reconstructing its input instead of forecasting a class label. This lesson details the compression of Fashion-MNIST images via an encoder, their passage through a reduced bottleneck representation, and subsequent reconstruction by a decoder. The diminishing reconstruction loss



indicated that the model progressively learned to provide outputs more akin to the original images. The visual comparisons indicated that the reconstructed images retained the primary forms of clothing items, but with diminished fine details. The bottleneck experiment yielded the most explicit lesson. A bottleneck size of 8 resulted in enhanced compression and increased error, but sizes 32 and 64 yielded crisper reconstructions with reduced error. Nevertheless, the enhancement from 32 to 64 was minimal, indicating declining returns. The tutorial demonstrates that the quality of reconstruction is contingent upon the extent of valuable information the bottleneck permits the model to retain and convey to the decoder. Autoencoders are effective for acquiring concise representations of images in practical applications.

## References

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